WeCloudData

Transformers

Introduction

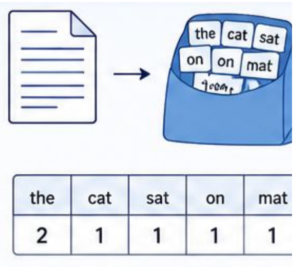
Introduction
Transformer

Why Did We Need Transformers?

Introduction

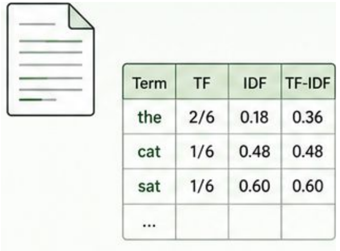
Before transformers, NLP relied on:

Bag of Words



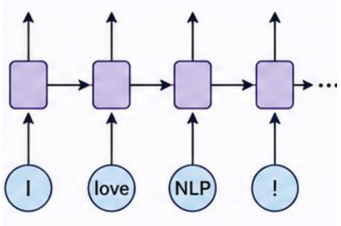
the	cat	sat	on	mat
2	1	1	1	1

TF-IDF

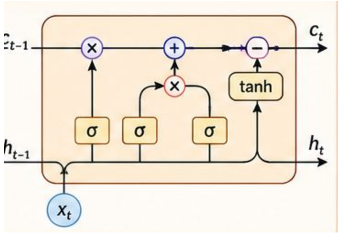


Term	TF	IDF	TF-IDF
the	2/6	0.18	0.36
cat	1/6	0.48	0.48
sat	1/6	0.60	0.60
...			

RNN



LSTM



Why do We Need Transformers?

Introduction

Those models struggled with:

Long Documents

Hard to capture meaning across many sentences

Long-Term Dependencies

Struggles to remember information far back in sequence.

Slow Training

Sequential processing makes training time consuming

Limited Parallelization

Most computations depend upon previous steps.

These limitations led to the need for a better approach Transformers.

Why do We Need Transformers?

Example

"I grew up in France.

...

After ten years, I moved back to **Paris**."

Other models may forget:

France

Transformers connect it better.

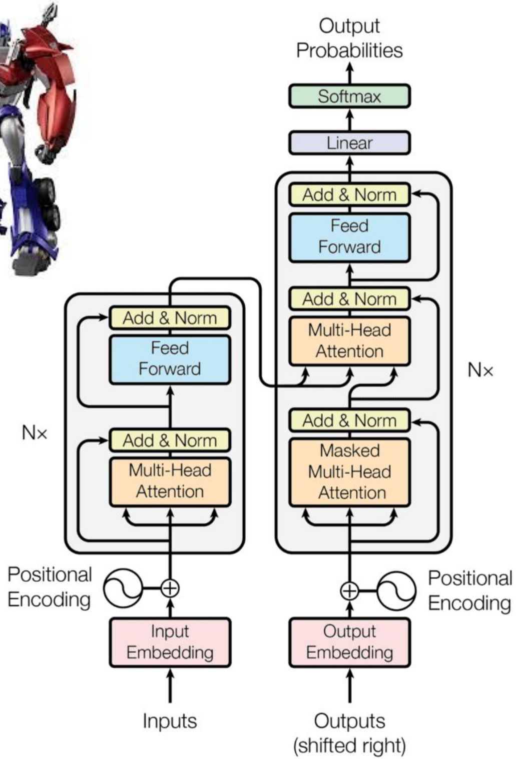
What is a Transformer?

Introduction

A Transformer is a deep learning architecture designed to process sequences efficiently.

Instead of relying on step-by-step recurrence,

it uses **attention** to connect words across the sequence.



https://arxiv.org/abs/1706.03762

Why do we need Transformers?

Introduction

Understand Long-range relationships

Faster Training

Better Language Understanding

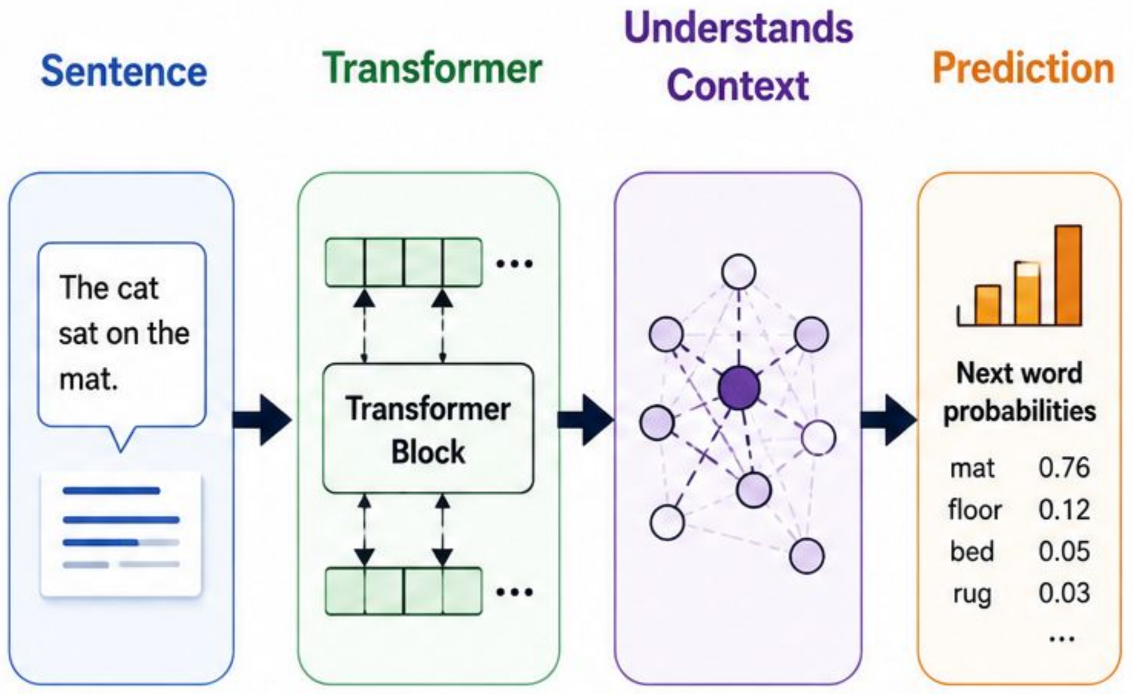
Scales to billions of Parameters

When can we use Transformers?

Introduction

How Transformers Work?

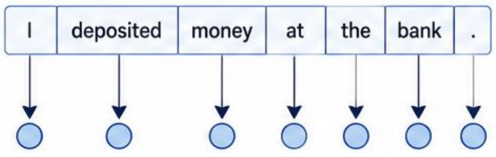
Introduction



Why Are Transformers So Powerful?

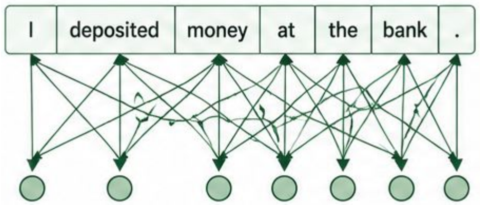
Introduction

Traditional NLP



- Mostly fixed representations
- One vector per word
- Limited sentence-level context
- Requires separate task models

Transformers



- Contextual representations
- One word, different meanings
- Connect words across the sentence
- Adapt to many language tasks

Why Are Transformers So Powerful?

Example

I deposited money in the bank

Vs

I sat on the river bank

Transformers understands

Same word, different meanings

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Attention

Transformer

What is Attention?

Attention

Imagine reading a sentence:

Do you give every word equal importance?

NO

You naturally focus on important words. Transformers do the same.

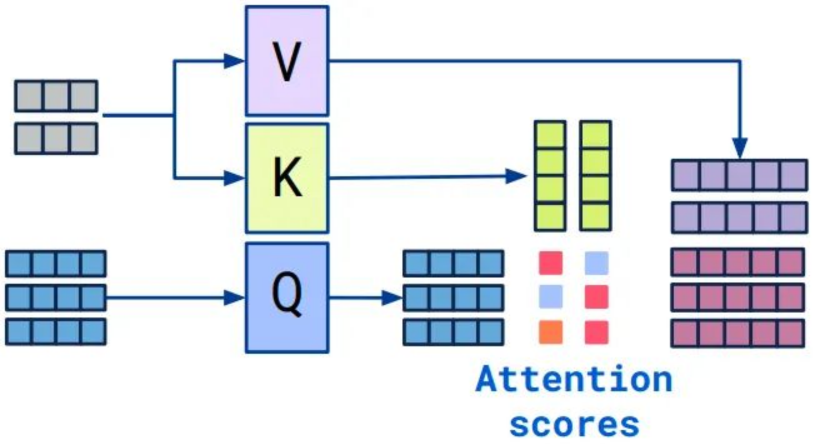
This mechanism is called:

Attention

What is Attention?

Attention

Attention allows a model to decide **which words deserve the most focus** while processing a sentence.



https://medium.com/@kalra.rakshit/introduction-to-transformers-and-attention-mechanisms-c29d252ea2c5

How Does Attention Work?

Attention

Sentence

The cat drank the milk because **it** was thirsty

Question

What does “it” refer to?

Attention helps connect

it ⟹ **cat**

Instead of

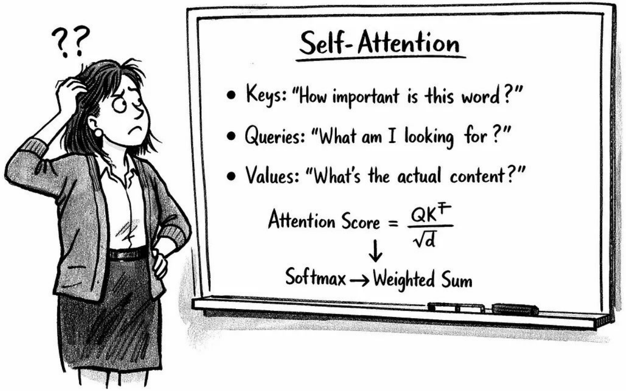
milk

What is Self-Attention?

Attention

Each word looks at the others and asks:
"Which words matter to me?"

This builds a **context-aware representation.**

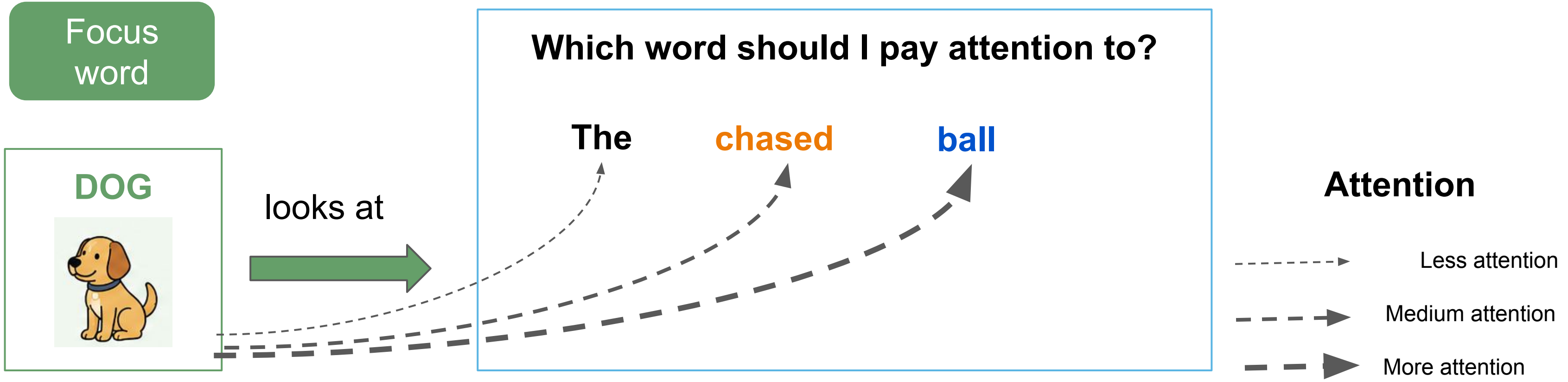


https://alexiskirke.medium.com/self-attention-finally-a-description-we-can-understand-intuitively-dbacf1f66a79

What is Self-Attention?

Example

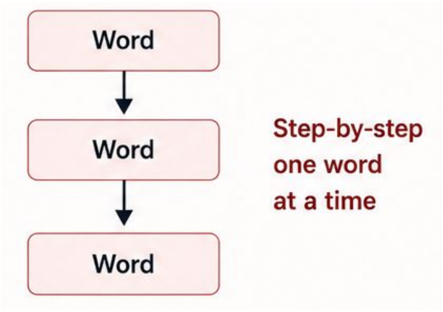
The dog chased the ball.



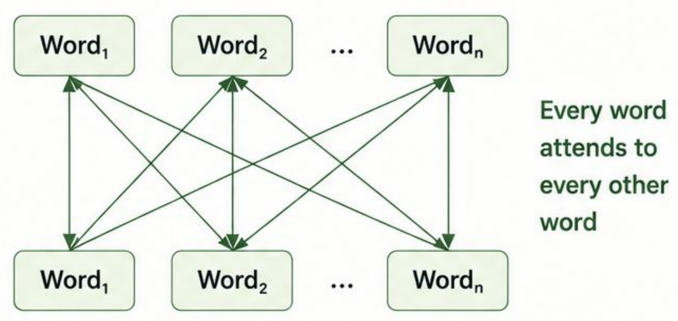
Why Attention Changed NLP?

Attention

Before Attention



After Attention



Why Attention Changed NLP?

Attention

Before Attention

Sequential
Slow
Forgets context

After Attention

Parallel
Fast
Better context

Encoder Transformer

What is an Encoder?

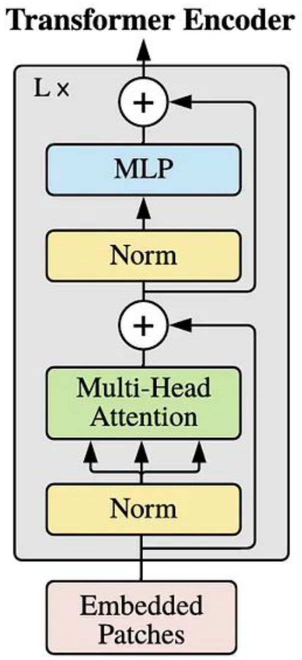
Encoder

The encoder reads the entire input sentence

and creates a rich contextual representation.

Think of it as

the reader



https://medium.com/@RobuRishabh/types-of-transformer-model-1b52381fa719

Why and When are Encoders Used?

Encoder

Why?

- Understand meaning
- Capture context
- Prepare information

When

Tasks requiring understanding

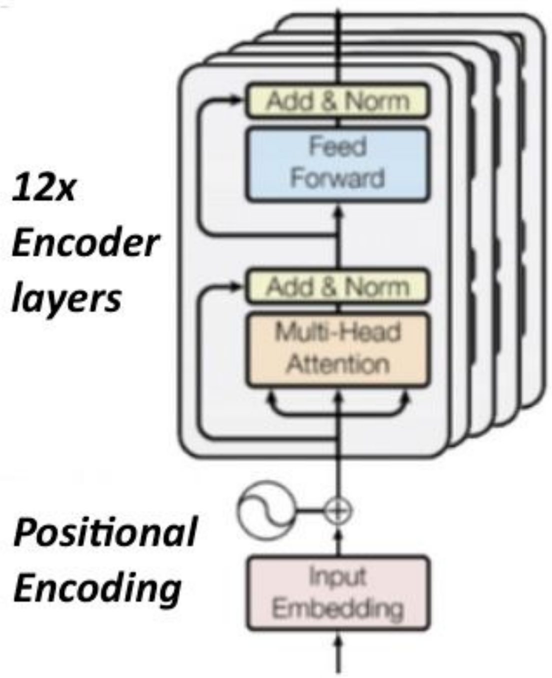
- Classification
- Sentiment Analysis
- Named Entity Recognition
- Search
- Question Answering

BERT (Encoder-only Model)

Encoder

Bidirectional Encoder Representations from Transformers

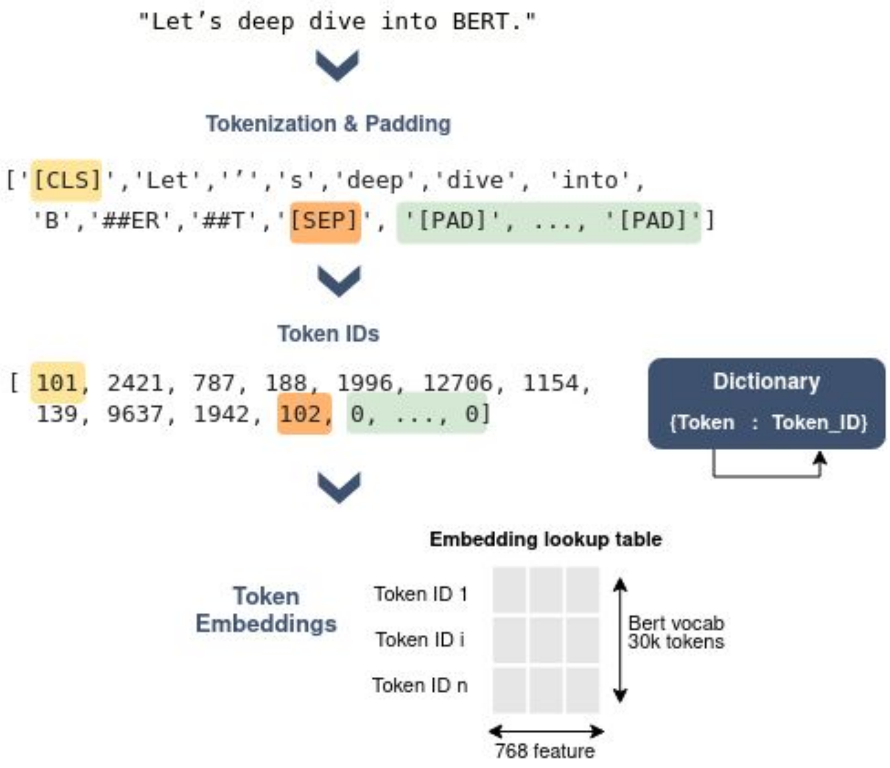
Uses context from both the left and the right simultaneously.

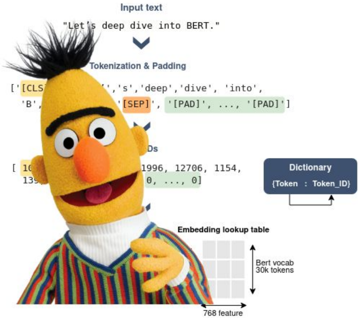


https://d3mlabs.de/?p=1169

BERT (Encoder Model)

Example

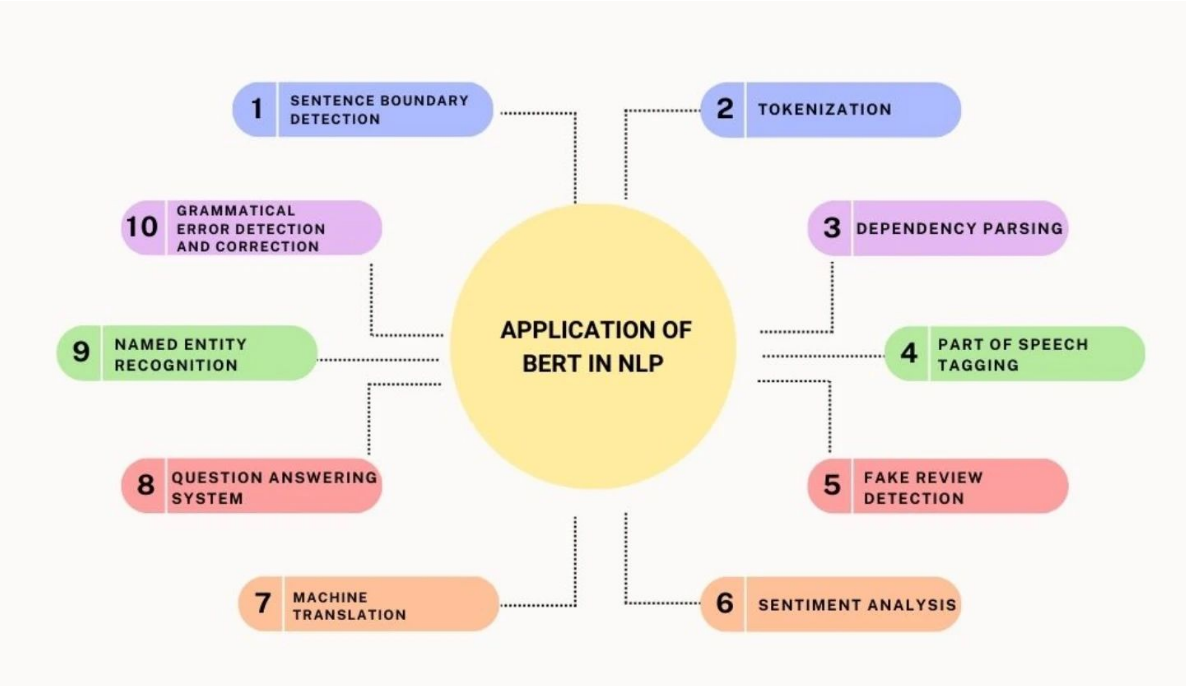




https://d3mlabs.de/?p=1169

Applications

Encoder



https://link.springer.com/article/10.1007/s10462-025-11162-5

Decoder Transformer

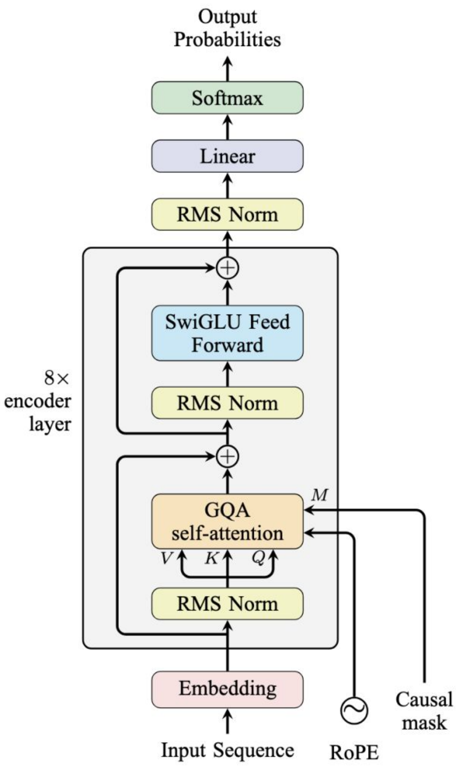
What is a Decoder?

Decoder

Think of the decoder as

the writer

A decoder uses the available context to predict the next token and generate an output sequence.



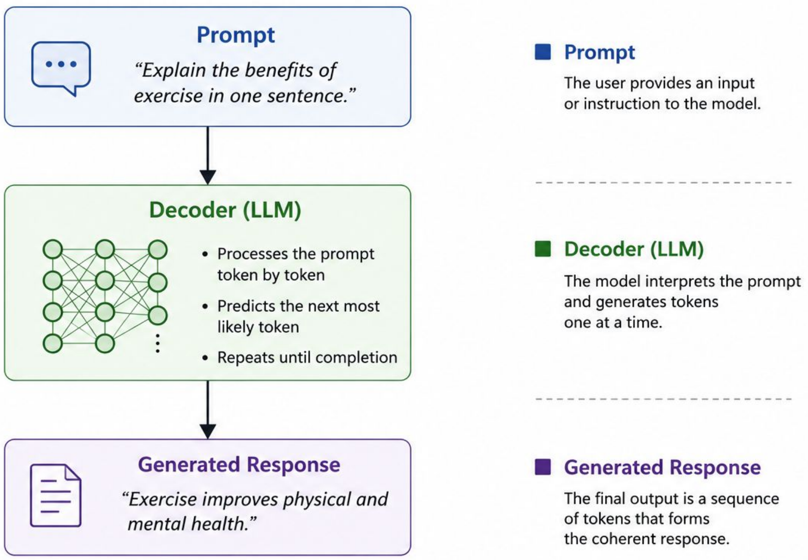
https://machinelearningmastery.com/building-a-decoder-only-transformer-model-for-text-generation/

When are Decoders used?

Decoder

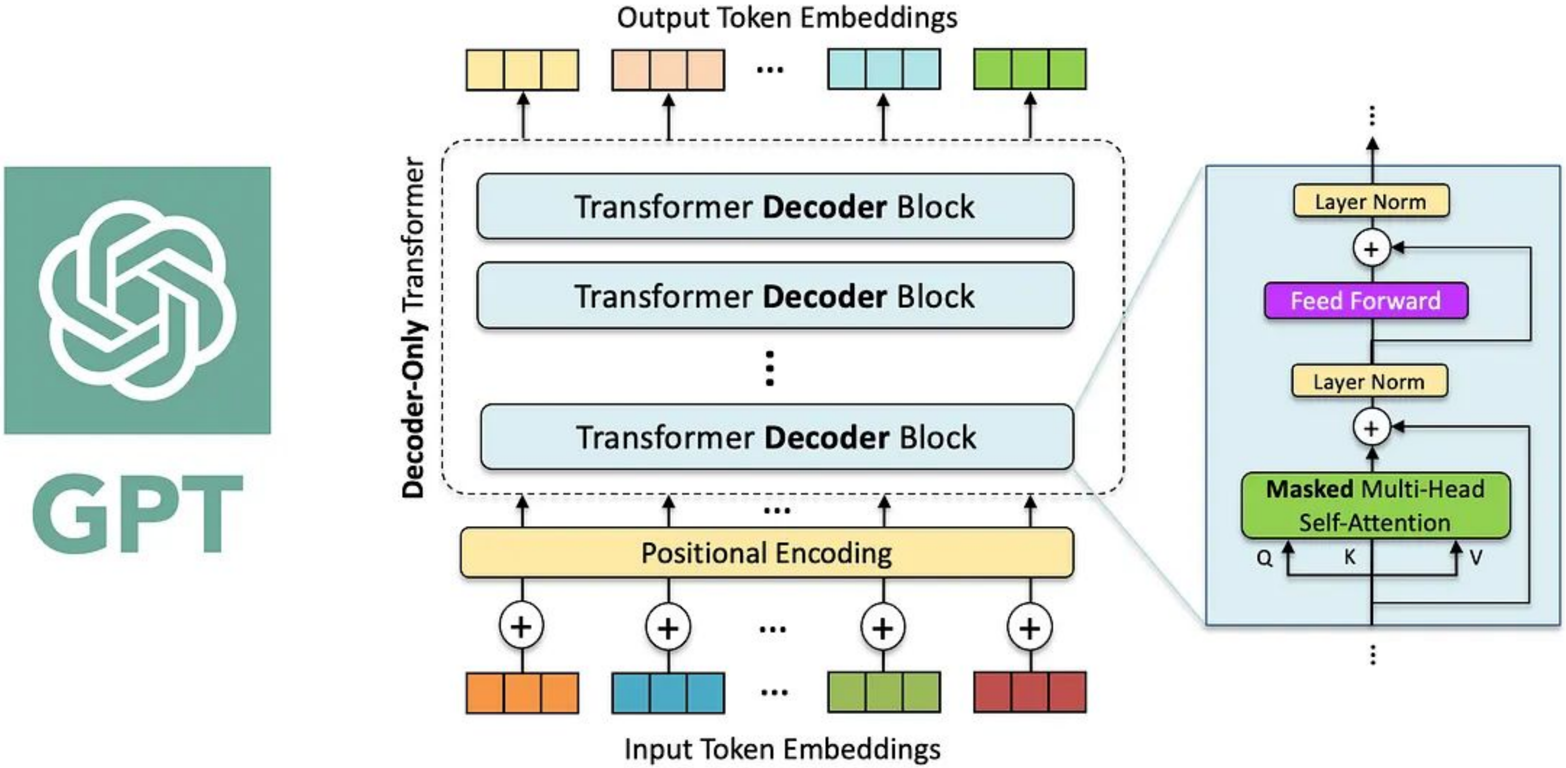
Why?

- Chatbots
- Story generation
- Code generation
- Translation
- Email writing



GPT (Decoder Model)

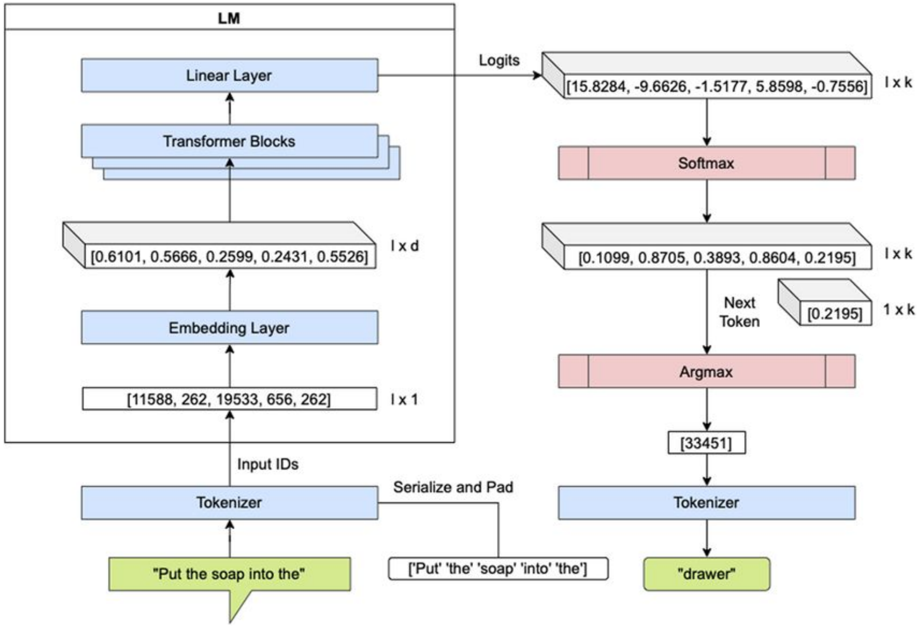
Decoder



https://medium.com/@lmpo/from-gpt-1-to-gpt-4o-1a901ac809d3

GPT (Decoder Model)

Example



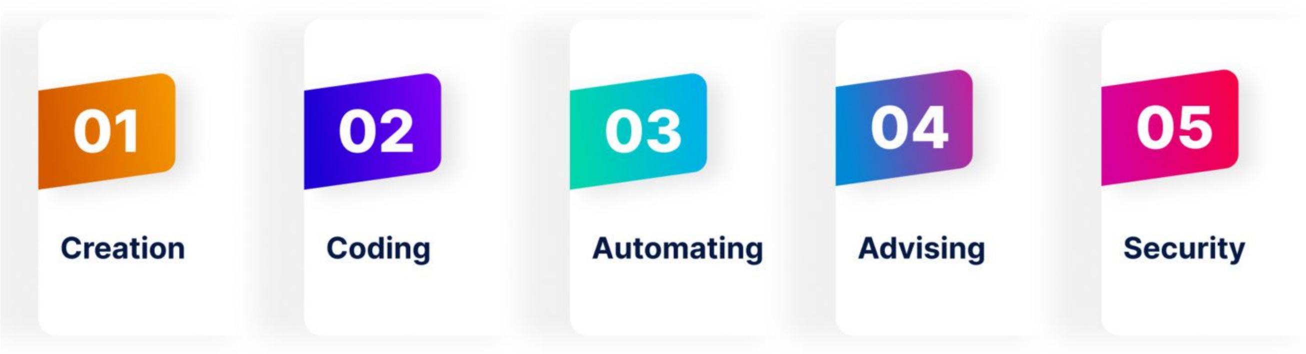
Unlike BERT, GPT cannot see future words.

It only uses the context **to its left.**

https://www.researchgate.net/publication/373183262_Learning_to_reason_over_scene_graphs_a_case_study_of_finetuning_GPT-2_into_a_robot_language_model_for_grounded_task_planning

Applications

Decoder



https://www.solulab.com/real-world-applications-of-generative-ai-and-gpt/

Encoder vs Decoder

Encoder Vs Decoder

Encoder	Decoder
Reads entire sentence	Generate Sentences
Bidirectional	Left to Right
Understand	Create
Classification	Generation
Search	Chatbot

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Pretrained Models
Transformer

What are Pre-trained Models?

Pretrained Models

What?

Models already trained on enormous text datasets.

Why?

No need to train from scratch. Saves

- Time
- Money
- Compute

When?

Almost every NLP project today.

What are Pre-trained Models?

Examples

https://computing4all.com/large-language-model-llm/

Why Use Pre-trained Models?

Pretrained Models

Training

Months

↓

Millions of dollars

↓

Thousands of GPUs

Using Pretrained Models

Download

↓

Fine-tune

↓

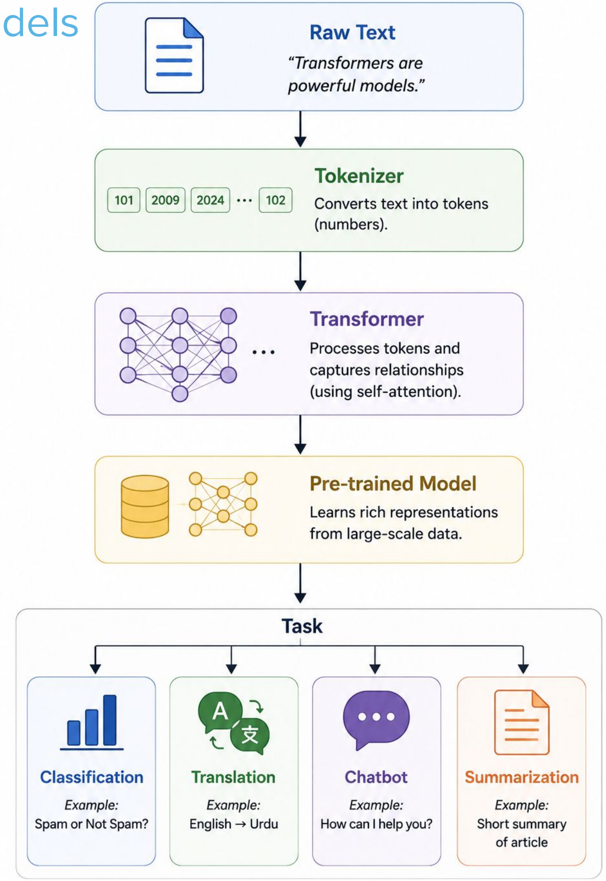
Deploy

Train from scratch

Download pre-trained model

How Modern NLP Works

Pretrained Models



Industry Examples

Pretrained Models

TASK	MODEL
Search	BERT
Chatbot	GPT
Translation	T5
Coding	Code Llama
Document QA	BERT + Retrieval

Summary

- Understood why Transformers replaced traditional NLP models.
- Learned what a Transformer is and why it became the foundation of modern LLMs.
- Explored the concept of Attention and Self-Attention for capturing contextual relationships.
- Learned how Encoder models (such as BERT) understand text by processing the entire input.
- Learned how Decoder models (such as GPT) generate text one token at a time.
- Compared Encoder and Decoder architectures and identified their ideal use cases.
- Understood the importance of pre-trained models and why they save significant time and computational resources.
- Explored real-world applications of Transformer-based models across modern NLP systems.

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Thank You

We'd love to hear from you. If you have any questions, feel free to reach out to our team anytime!